A STANDARD FPQC COVERING IS A STANDARD V COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3625, 3628] and proof(s) [[3630, 3656]]; this spec covers the complete environment instead (statement 3625-3628, proof span(s) [(3630, 3656)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original definition 022F: sources/stacks-chow/tags.txt, source lines 2704–2704. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01J8: sources/stacks-chow/tags.txt, source lines 2005–2005. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *A standard fpqc covering is a standard V covering.*

Proof. Let $\{X_i \to X\}_{i=1,\ldots,n}$ be a standard fpqc covering (Definition 022F). Let $g : \mathrm{Spec}(V) \to X$ be a morphism where V is a valuation ring. Let $x \in X$ be the image of the closed point of $\mathrm{Spec}(V)$. Choose an i and a point $x_i \in X_i$ mapping to x. Then $\mathrm{Spec}(V) \times_X X_i$ has a point x_i' mapping to the closed point of $\mathrm{Spec}(V)$. Since $\mathrm{Spec}(V) \times_X X_i \to \mathrm{Spec}(V)$ is flat we can find a specialization $x_i'' \rightsquigarrow x_i'$ of points of $\mathrm{Spec}(V) \times_X X_i$ with x_i'' mapping to the generic point of $\mathrm{Spec}(V)$, see Morphisms, Lemma 03HV. By Schemes, Lemma 01J8 we can choose a valuation ring W and a morphism $h : \mathrm{Spec}(W) \to \mathrm{Spec}(V) \times_X X_i$ such that h maps the generic point of $\mathrm{Spec}(W)$ to x_i'' and the closed point of $\mathrm{Spec}(W)$ to x_i'. We obtain a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(W) & \longrightarrow & X_i \\ \downarrow & & \downarrow \\ \mathrm{Spec}(V) & \longrightarrow & X \end{array}$$

where $V \to W$ is an extension of valuation rings. This proves the lemma. □

Date: stacks-project:topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.